

BALTIMORE HARBOR, MD, USAWMO: **997296**Lat: **39.2669N**Lon: **76.5794W**Elev: **33**StdP: **14.68**Time Zone: **-5.00 (NAE)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.0	19.3	N/A	N/A	N/A	N/A	N/A	N/A	20.1	27.5	18.5	29.5	8.3	330	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	92.6	N/A	90.1	N/A	87.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.2	16.2	13.8	11.0	97.3	5.2	3.6	7.3	99.9	4.2	102.0	1.3	104.1	-2.6	106.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	58.0	34.1	37.3	44.5	57.2	67.5	75.5	80.1	77.4	71.4	60.0	48.6	41.4	(6)
		DBStd	17.33	8.94	8.77	8.84	7.87	7.14	5.57	4.75	3.80	5.75	7.04	7.42	8.04	
		HDD50	1509	497	367	218	21	0	0	0	0	9	109	288		
		HDD65	4109	959	777	637	257	58	3	0	0	9	186	491	732	
		CDD50	4437	3	10	47	238	542	764	934	848	643	318	68	22	
		CDD65	1560	0	0	1	24	135	317	469	383	201	30	0	0	
		CDH74	12783	0	2	8	199	807	2600	4810	3034	1195	128	0	0	
		CDH80	4121	0	0	0	63	198	823	1900	825	290	22	0	0	
(14)	Wind	WSAvg	6.0	6.3	6.4	7.0	6.5	5.8	5.1	5.3	5.8	6.1	6.1	5.9		(14)
(15)	Precipitation	PrecAvg	42.0	3.1	3.1	3.9	3.1	3.9	3.1	3.8	4.1	3.6	3.0	3.5	3.7	(15)
(16)		PrecMax	61.6	8.7	6.9	9.8	7.6	9.3	10.0	11.7	14.8	8.8	8.7	8.7	8.1	(16)
(17)		PrecMin	29.6	0.4	0.7	0.9	0.4	0.6	0.6	0.7	0.3	0.1	0.1	0.4	0.7	(17)
(18)		PrecStd	8.8	1.7	1.7	2.1	1.7	2.0	1.8	2.1	3.0	2.4	1.9	1.9	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	60.2	68.5	73.9	87.6	89.2	94.2	97.8	92.0	90.4	82.4	70.3	66.9		(19)
(20)		0.4% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(20)
(21)		2% DB	55.1	60.1	67.5	79.7	85.3	90.8	94.5	88.9	86.6	77.5	65.7	61.3		(21)
(22)		2% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(22)
(23)		5% DB	50.2	53.4	62.5	73.7	81.7	87.6	91.4	86.7	83.4	73.4	62.4	57.6		(23)
(24)		5% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(24)
(25)		10% DB	46.5	48.8	57.7	69.1	78.5	84.7	88.5	84.6	80.4	70.4	59.5	53.5		(25)
(26)		10% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)
(28)		0.4% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)		2% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)		5% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)		10% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range	MDBR	12.2	12.7	13.5	15.5	13.9	13.0	13.1	12.1	12.0	12.6	12.5	11.2		(35)
(36)		5% DB MCDBR	18.5	20.5	21.2	22.6	17.5	16.7	16.1	14.5	14.8	16.6	16.5	16.8		(36)
(37)		5% DB MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(37)
(38)		5% WB MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(38)
(39)		5% WB MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.322	0.362	0.401	0.446	0.500	0.518	0.513	0.429	0.364	0.326	0.311		(40)
(41)		taud	2.467	2.422	2.374	2.287	2.194	2.085	2.050	2.048	2.255	2.443	2.507	2.507		(41)
(42)		Ebn at Noon	273	286	283	278	266	251	245	242	258	266	266	265		(42)
(43)		Edh at Noon	25	30	35	41	46	51	53	51	39	30	24	23		(43)
(44)	All-Sky Solar Radiation	RadAvg	643	914	1212	1557	1741	1932	1913	1693	1380	1020	735	563		(44)
(45)		RadStd	53	106	90	126	209	149	94	96	162	111	62	55		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (79 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+232	+131

Nomenclature: See separate page